

Lab 03

spice

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5 Measure the FO4 delay for the middle inverter in the previous testbench. Calculate FO4 delay (in ps) divided by lambda (in nm), where $\lambda = L_{min} / 2$. This ratio is useful to get a quick estimate of the delay in any technology. How much FO4 delay do you expect for a 65nm technology?21

Part 1 (prelab)

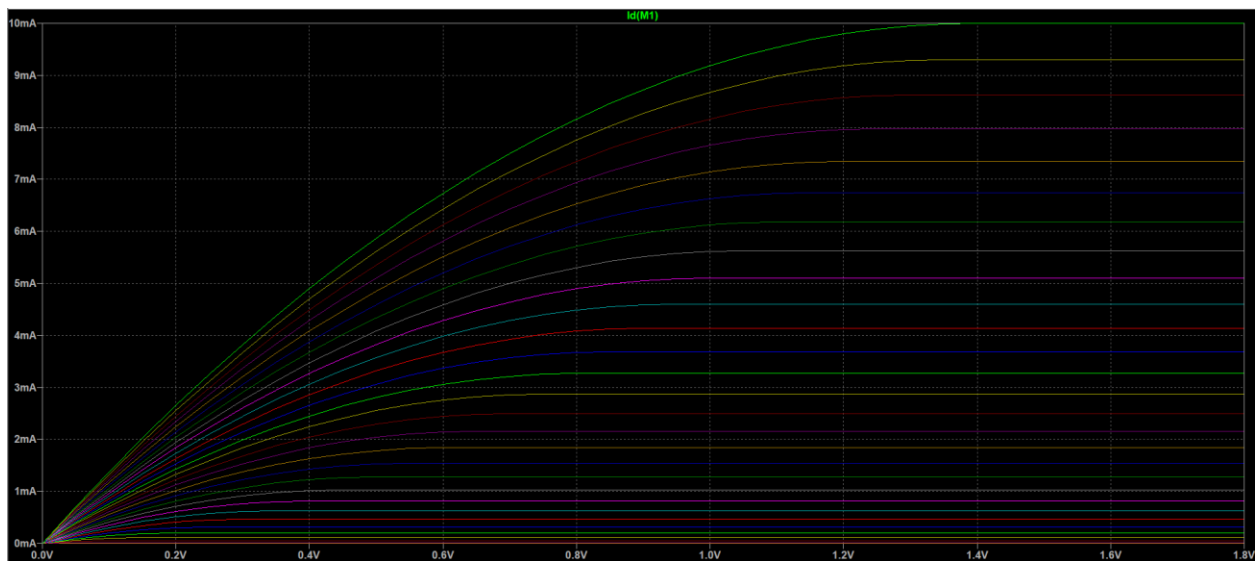
1 COMPLETE “MY_NMOS.LIB”, COMPLETE AND RUN “NMOS_TB.CIR”

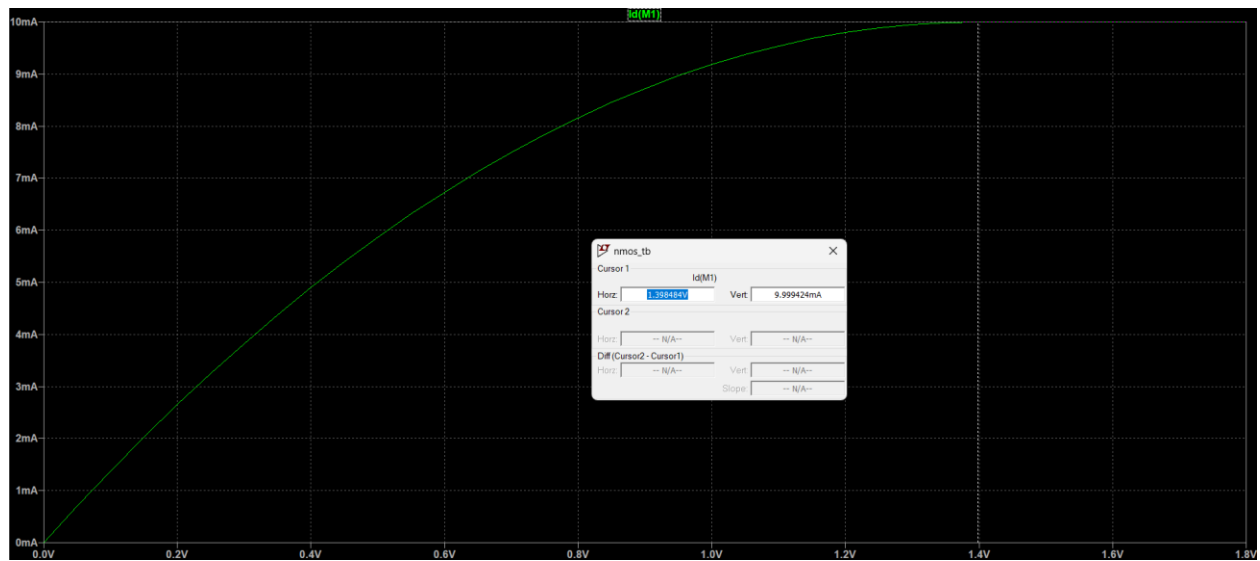
```
* Part 1.1
.MODEL nmos_part1_1 NMOS [level=1 Vto=0.4 kp=183.67u]
```

$$I = \frac{1}{2} K_P \frac{W}{L} (V_{GS} - V_{TH0})^2$$

So $k_p = 183.67 \text{ } \mu\text{m}$

```
NMOS testbench
.include my_nmos.lib
** Circuit Description **
VGS 1 0 DC 1.8V
VDS 2 0 DC 1.8V
M1 2 1 0 0 nmos_part1_1 W=10u L=180n
** Analysis Requests **
.DC VDS 0 1.8 0.05 VGS 0 1.8 0.05
.END
```





LTspice 24.0.12 for Windows

Circuit: NMOS testbench

Start Time: Tue Jul 30 23:23:29 2024

solver = Normal

Maximum thread count: 12

tnom = 27

temp = 27

method = modified trap

Instance "m1": Length shorter than recommended for a level 1 MOSFET.

Instance "m1": Width narrower than recommended for a level 1 MOSFET.

.OP point found by inspection.

.step vgs=0

step vgs=0 0.5

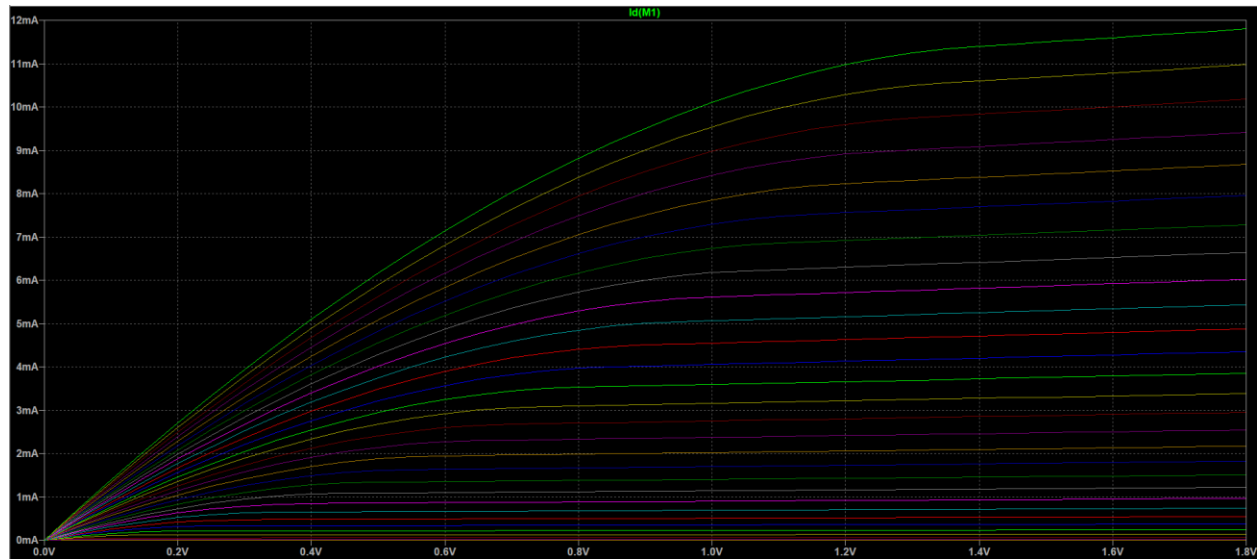
2 MODIFY THE MODEL TO INCLUDE VDS DEPENDENCE WITH $V_A = 10\text{ V}$. REPORT THE NETLIST AND THE RESULTS.

★ Part1.2

★

```
.MODEL nmos_part1_2 NMOS [level=1 Vto=0.4 kp=183.67u Lambda=0.1]
```

★ Don't forget to change the model name to nmos_part1_2



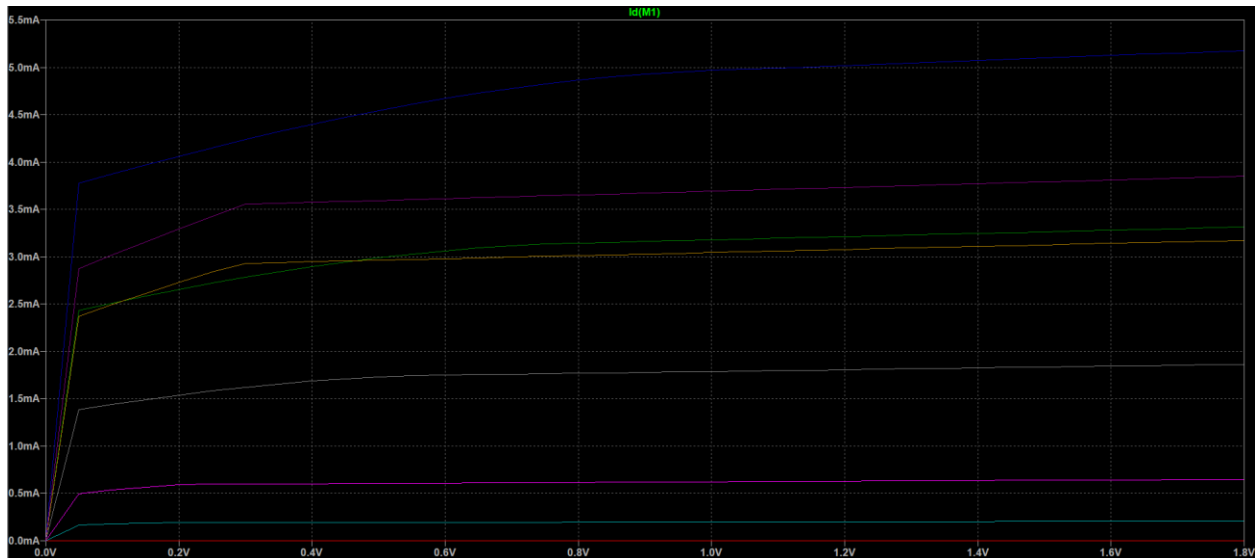
3 MODIFY THE MODEL TO INCLUDE THE EFFECT OF VELOCITY SATURATION THAT LIMITS THE ON CURRENT TO 5 mA. REPORT THE NETLIST AND THE RESULTS

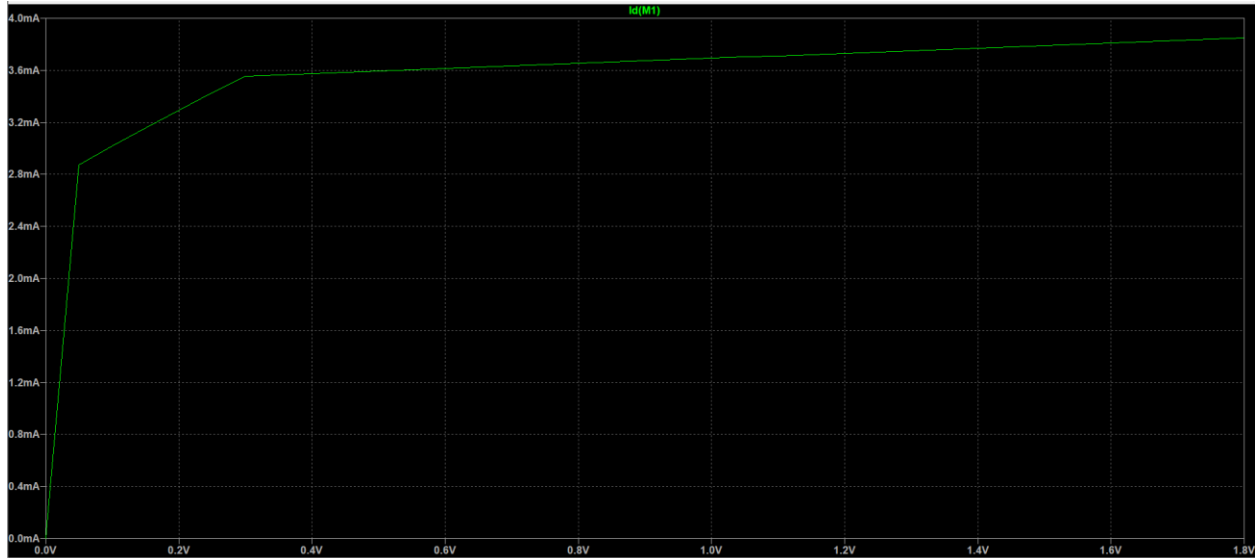
$$I_D = C_{ox} W v_{sat} (V_{ov} - \frac{V_{DS_{SAT}}}{2}) (1 + \lambda V_{ds})$$

$$I_D = \frac{Kp}{u} W v_{sat} (V_{OV} - \frac{LV_{sat}}{2u})$$

$$v_{sat} = 112.410k$$

```
.MODEL nmos_part1_3 NMOS [LEVEL=2 VTO=0.4 KP=183.67u LAMBDA=0.1 VMAX=112.410k]
```

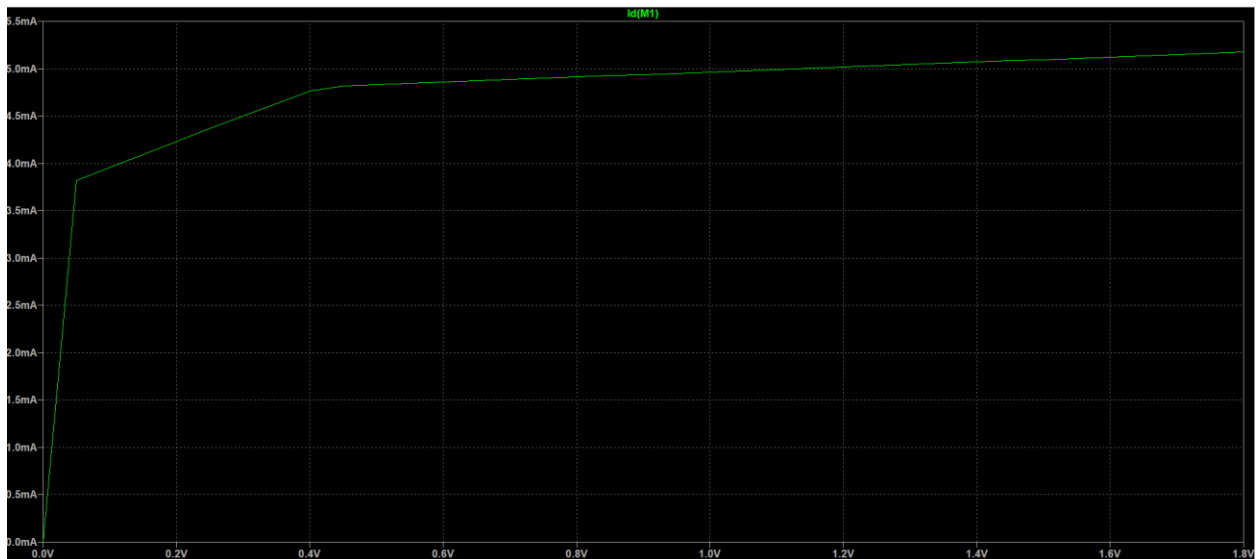




At $v_{gs}=1.8V$

If we used level 3 and did some iterations we would reach the exact required.

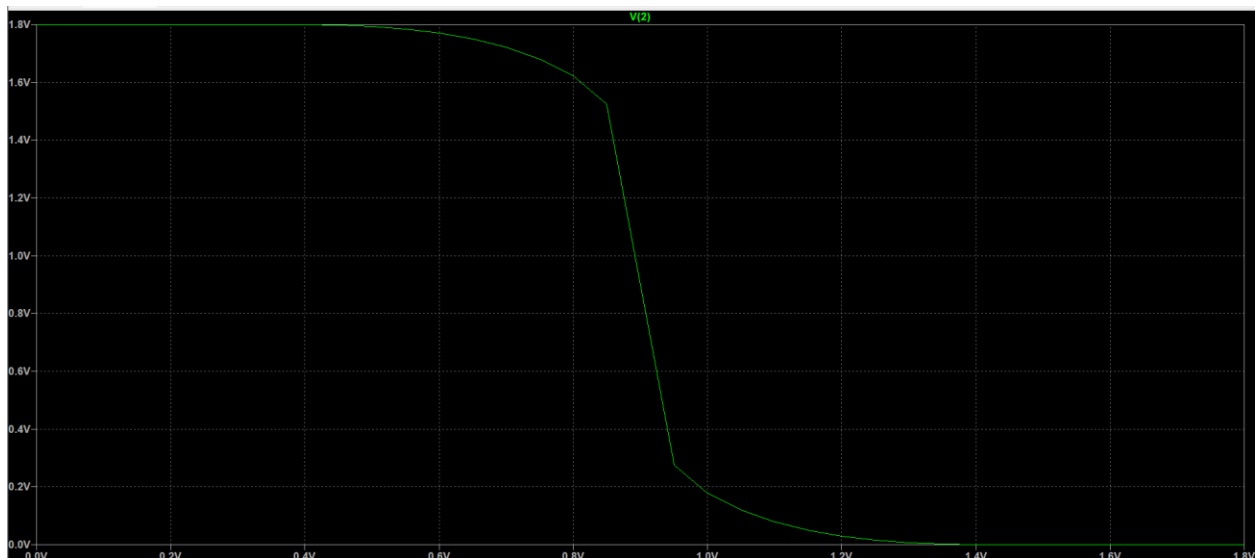
```
.MODEL nmos_part1_3 NMOS [level=2 Vto=0.4 kp=183.67u Lambda=0.1 VMAX=170e3]
```



4 CREATE "MY_PNOS.LIB" TO MODEL A LEVEL 1 PMOS THAT HAS HALF THE MOBILITY OF THE NMOS (USE THE NMOS MODEL IN STEP 2). WHAT SHOULD BE THE POLARITY OF VT0? COMPLETE AND RUN "CMOS_INV_DC.CIR". THIS SHOULD PLOT THE DC TRANSFER CHARACTERISTICS OF A CMOS INVERTER. REPORT THE NETLIST AND THE RESULTS.

```
.MODEL pmos_part1_4 PMOS [level=1 Vto=-0.4 kp=91.835u Lambda=0.1]
```

Vt0 should be -ve as it has the opposite convention of nmos, as the gate needs to be at lower voltage than source for channel to be formed.



```

CMOS inverter DC sweep
** Circuit Description **
* power supply
VDD 3 0 DC 1.8V
* input
VIN 1 0 DC 0V
* circuit
M1 2 1 3 3 pmos_part1_4 W=20u L=180n
M2 2 1 0 0 nmos_part1_2 W=10u L=180n
.include my_nmos.lib
.include my_pmos.lib
** Analysis Requests **
.DC VIN 0 1.8 0.05
** Outputs Requests **
.probe V(2)
.END

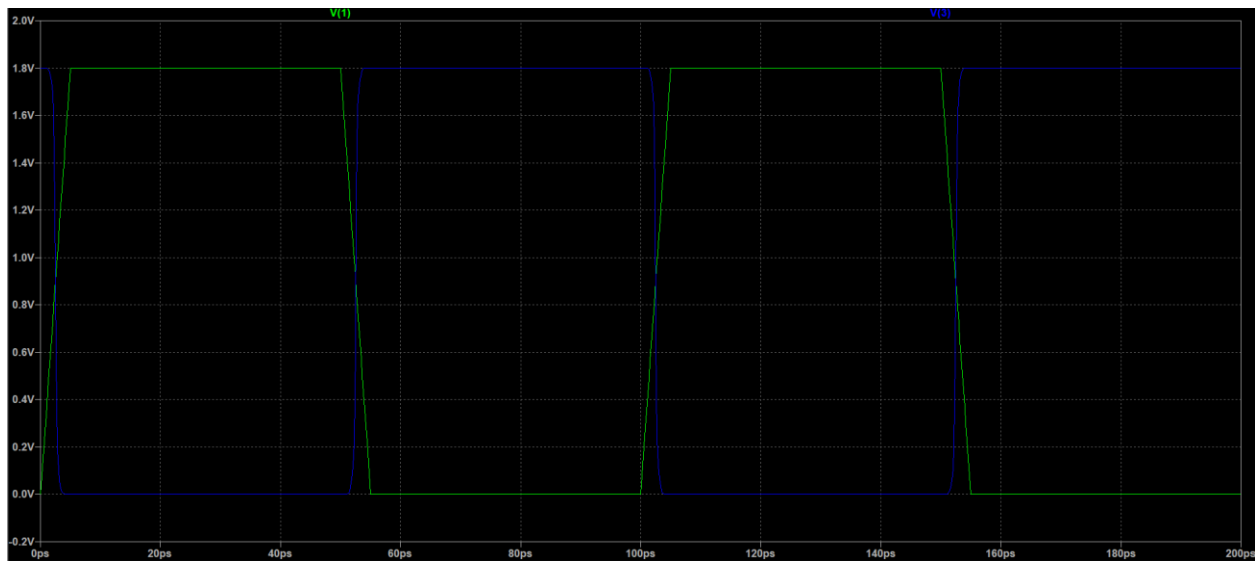
```

- 5 COMPLETE AND RUN “CMOS_INV_TRAN.CIR”. THIS SHOULD PLOT THE TRANSIENT CHARACTERISTICS OF A CMOS INVERTER. USE PULSE SOURCE WITH PERIOD = 100 PS AND RISE/FALL = 5 PS. REPORT THE NETLIST AND THE RESULTS. MEASURE THE DELAY

```

CMOS inverter tran analysis
.include my_nmos.lib
.include my_pmos.lib
** Circuit Description **
VDD 2 0 DC 1.8V
Vin 1 0 PULSE(0 1.8 0 5p 5p 45p 100p)
M1 3 1 2 2 pmos_part1_4 W=20u L=180n
M2 3 1 0 0 nmos_part1_2 W=10u L=180n
#M1 3 1 2 2 pmos_part1_6 W=20u L=180n AD=9p AS=9p PD=40.9u PS=40.9u
#M2 3 1 0 0 nmos_part1_6 W=10u L=180n AD=4.5p AS=4.5p PD=20.9u PS=20.9u
** Analysis Requests **
.TRAN 1p 200p
*delay defined from 50% of input to 50% of output
.MEAS TRAN Tphl TRIG when v(1) =0.9 cross=1 TARG when v(3)=0.9 cross=1
.MEAS TRAN Tplh TRIG when v(1) =0.9 cross=2 TARG when v(3)=0.9 cross=2
.END

```



 SPICE Output Log: D:\spice\LAB3\cmos_inv_tran.log

```

LTspice 24.0.12 for Windows
Circuit: CMOS inverter tran analysis
Start Time: Wed Aug  7 01:33:33 2024
solver = Normal
Maximum thread count: 12
tnom = 27
temp = 27
method = modified trap
Instance "m1": Length shorter than recommended for a level 1 MOSFET.
Instance "m2": Length shorter than recommended for a level 1 MOSFET.
Instance "m2": Width narrower than recommended for a level 1 MOSFET.
Direct Newton iteration for .op point succeeded.

tphl=1.4684e-15 FROM 2.5e-12 TO 2.50147e-12
tplh=2.57719e-14 FROM 5.25e-11 TO 5.25258e-11

Total elapsed time: 0.164 seconds.

```

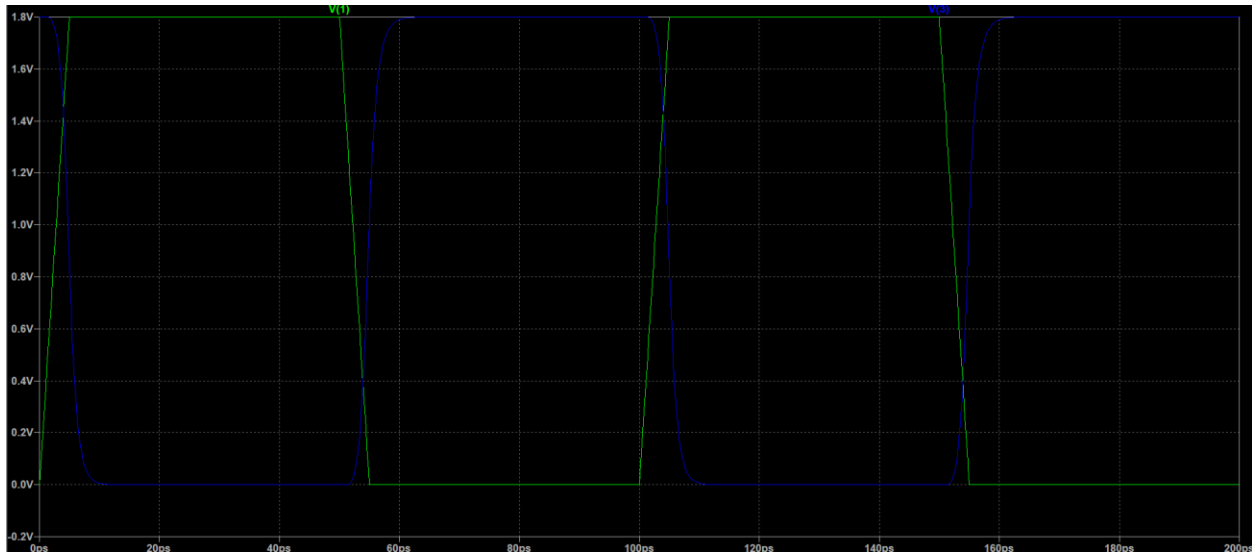
$$\text{Delay} = \frac{(t_{phl} + t_{plh})}{2} = 13.6\text{fs}$$

Ideally it's 0 so it just numerical error from spice.

6 MODIFY THE MODEL TO INCLUDE $CJ = 1 \text{ fF}/(\text{UM})^2$ AND $CJSW = 0.1 \text{ fF}/\text{UM}$ (NOTE THE UNITS). USE REASONABLE VALUES FOR (AD,AS,PD,PS) WHEN YOU INSTANTIATE THE MOSFET. PAGE 2 OF 2 ASSUME DIFFUSION (JUNCTION) LENGTH (LDIFF) = $5 * \text{LAMBDA}$, $\text{LAMBDA} = \text{LMIN}/2$, JUNCTION AREA = $\text{LDIFF} * W$ AND JUNCTION PERIMETER = $2 * (\text{LDIFF} + W)$. RUN “CMOS_INV_TRAN.CIR”. REPORT THE NETLIST AND THE RESULTS. MEASURE THE DELAY. COMPARE THE RESULTS TO THE PREVIOUS CASE. COMMENT.

```
.MODEL nmos_part1_6 NMOS [LEVEL=1 VTO=0.4 KP=183.67u LAMBDA=0.1 CJ=1e-3 CJSW=1e-10]  
*-----*
```

```
CMOS inverter tran analysis  
.include my_nmos.lib  
.include my_pmos.lib  
** Circuit Description **  
VDD 2 0 DC 1.8V  
Vin 1 0 PULSE(0 1.8 0 5p 5p 45p 100p)  
#M1 3 1 2 2 pmos_part1_4 W=20u L=180n  
#M2 3 1 0 0 nmos_part1_2 W=10u L=180n  
M1 3 1 2 2 pmos_part1_6 W=20u L=180n AD=9p AS=9p PD=40.9u PS=40.9u  
M2 3 1 0 0 nmos_part1_6 W=10u L=180n AD=4.5p AS=4.5p PD=20.9u PS=20.9u  
** Analysis Requests **  
.TRAN 1p 200p  
*delay defined from 50% of input to 50% of output  
.MEAS TRAN Tphl TRIG when v(1) =0.9 cross=1 TARG when v(3)=0.9 cross=1  
.MEAS TRAN Tplh TRIG when v(1) =0.9 cross=2 TARG when v(3)=0.9 cross=2  
.END
```



 SPICE Output Log: D:\spice\LAB3\cmos_inv_tran.log

```

LTspice 24.0.12 for Windows
Circuit: CMOS inverter tran analysis
Start Time: Wed Aug  7 01:34:42 2024
solver = Normal
Maximum thread count: 12
tnom = 27
temp = 27
method = modified trap
Instance "m1": Length shorter than recommended for a level 1 MOSFET.
Instance "m2": Length shorter than recommended for a level 1 MOSFET.
Instance "m2": Width narrower than recommended for a level 1 MOSFET.
Direct Newton iteration for .op point succeeded.

tphl=2.41003e-12 FROM 2.5e-12 TO 4.91003e-12
tplh=2.2999e-12 FROM 5.25e-11 TO 5.47999e-11

Total elapsed time: 0.202 seconds.

```

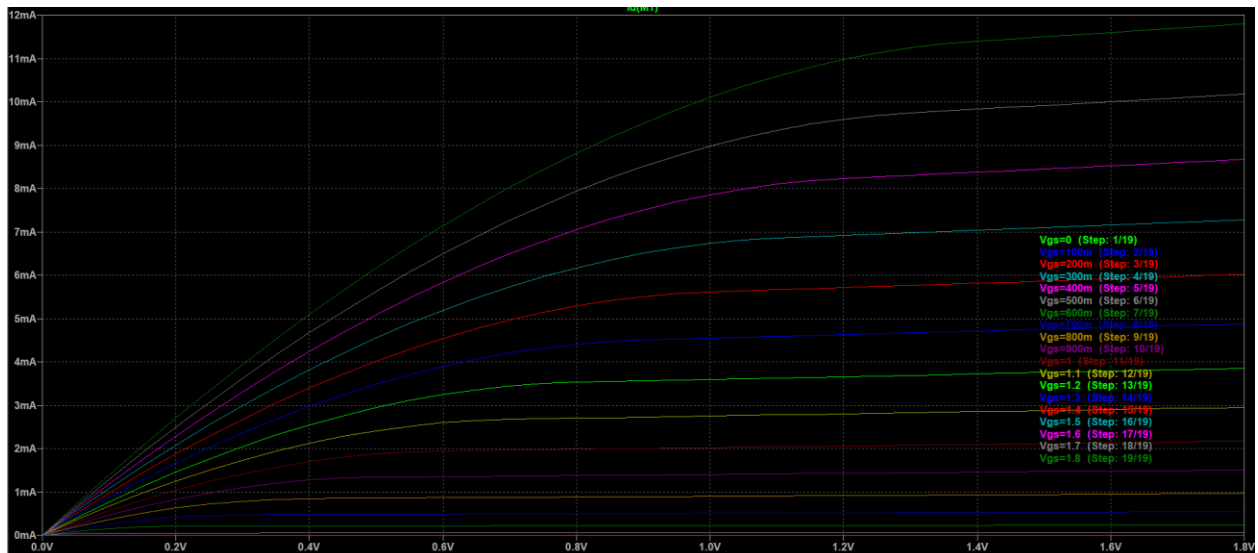
$$T_{delay} = \frac{t_{phl} + t_{plh}}{2} = 2.355ps$$

The delay is much larger than the previous case as expected, we introduced diffusion areas and perimeters so the capacitances would be considered.

PART2

- 1 RUN PARAMETRIC SWEEPS FOR BOTH VGS AND VDS INSTEAD OF NESTED DC SWEEPS. COMPARE THE SIMULATION TIME TO THE NESTED DC SWEEP TESTBENCH

```
*****
NMOS testbench
.include my_nmos.lib
.param VGS=1.8V
.param VDS=1.8V
** Circuit Description **
VGS 1 0 DC 1.8V
VDS 2 0 DC 1.8V
M1 2 1 0 0 nmos_part1_2 W=10u L=180n
** Analysis Requests **
#.DC VDS 0 1.8 0.05 VGS 0 1.8 0.2
.DC VDS 0 1.8 0.05
.step VGS 0 1.8 0.1
.END
```



Parametric sweep	Dc sweep
Total elapsed time: 0.317 seconds.	Total elapsed time: 0.057 seconds.

Dc sweep is much faster than parametric sweep.

2 EDIT “NMOS_TB.CIR” TO USE THE “EE214B_HSPICE.SP” HSPICE MODEL.

2.1 WHAT VERSION OF BSIM IS THIS HSPICE MODEL?

Level 49 and 53 BSIM3v3 MOS Models

The Synopsys Level 49 and Level 53 models are based on the BSIM3v3 MOS model from UC Berkeley.

LEVEL 49 in Hspice is HSPICE-enhanced version of BSIM3v3

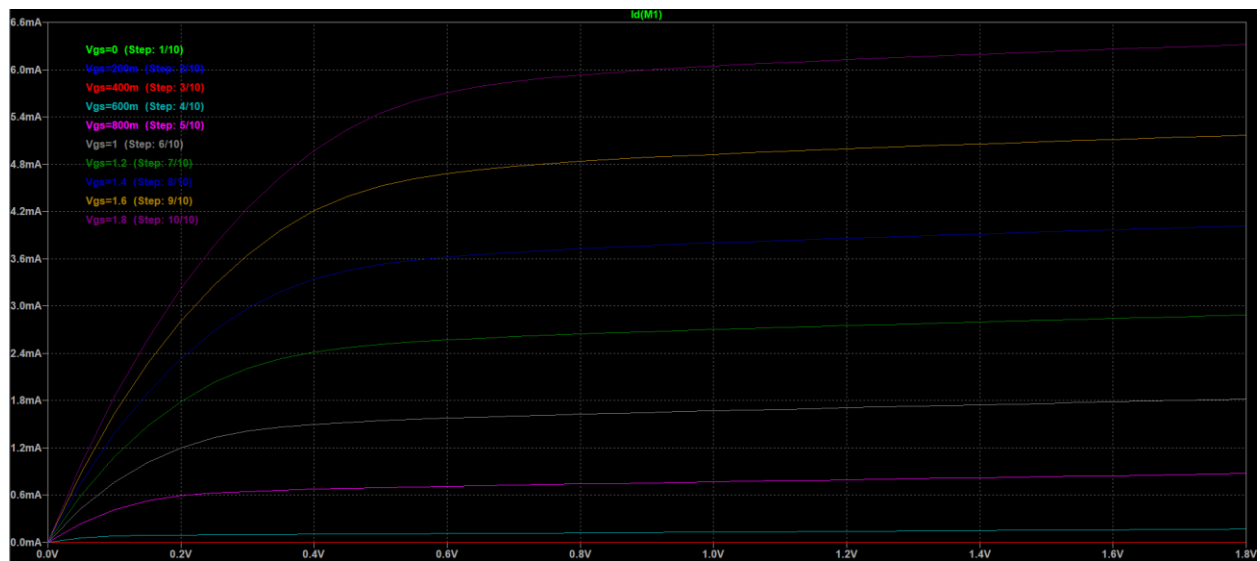
2.2 HOW TO SET THE LEVEL PARAMETER APPROPRIATELY FOR LTSPICE IN THIS CASE?

To be LEVEL 8

level model

- 1 Shichman-Hodges
- 2 MOS2(see A. Vladimirescu and S. Liu, The Simulation of MOS Integrated Circuits Using SPICE2, ERL Memo No. M80/7, Electronics Research Laboratory University of California, Berkeley, October 1980)
- 3 MOS3, a semi-empirical model(see reference for **level** 2)
- 4 BSIM (see B. J. Sheu, D. L. Scharfetter, and P. K. Ko, SPICE2 Implementation of BSIM. ERL Memo No. ERL M85/42, Electronics Research Laboratory University of California, Berkeley, May 1985)
- 5 BSIM2 (see Min-Chie Jeng, Design and Modeling of Deep-Submicrometer MOSFETs ERL Memo Nos. ERL M90/90, Electronics Research Laboratory University of California, Berkeley, October 1990)
- 6 MOS6 (see T. Sakurai and A. R. Newton, A Simple MOSFET Model for Circuit Analysis and its application to CMOS gate delay analysis and series-connected MOSFET Structure, ERL Memo No. ERL M90/19, Electronics Research Laboratory, University of California, Berkeley, March 1990)
- 8 BSIM3v3.3.0 from University of California, Berkeley as of July 29, 2005

2.3 COMPARE THE RESULTS TO YOUR LEVEL 1 MODEL.



The on current is 6mA, but in Level 1 was 10mA, steps between different V_{GS} is almost linear so it considers velocity saturation, also it considers μ_{eff} and dependence of V_{ds} .

3 WHEN YOU RUN THE PREVIOUS TESTBENCH, YOU WILL SEE IN THE SPICE ERROR LOG THAT LTSPICE HAD PROBLEMS WITH SOME BSIM PARAMETERS IN THE MODEL FILE. REFER TO HSPICE DOCUMENTATION AND EXPLAIN THE MEANING AND USE OF EACH PARAMETER.

```
SPICE Output Log: D:\spice\LAB3\nmos_tb.log
LTspice 24.0.12 for Windows
Circuit: NMOS testbench
Start Time: Mon Aug 5 12:44:08 2024
Ignoring BSIM parameter ACM
Ignoring BSIM parameter HDIF
Ignoring BSIM parameter XL
Ignoring BSIM parameter XW
Error on line 7 : .model nch nmos ( acm = 3 hdif = 0.32e-6 level = 8 version = 3.3 tnom = 27 tox = 4.1e-9 xj = 1e-7 nch = 2.3549e17 vth0 = 0.3618397 k1 =
* Unrecognized parameter "cjgate" -- ignored
solver = Normal
Maximum thread count: 12
tnom = 27
temp = 27
method = modified trap
Warning: Pd = 0 is less than W.
Warning: Ps = 0 is less than W.
.OP point found by inspection.
.step vgs=0
.step vgs=0.2
.step vgs=0.4
.step vgs=0.6
.step vgs=0.8
.step vgs=1
.step vgs=1.2
.step vgs=1.4
.step vgs=1.6
.step vgs=1.8
Total elapsed time: 0.586 seconds.
```

3.1 ACM:

ACM	-	10	No	Area calculation method selector (HSPICE specific)
• ACM=0-3 uses the HSPICE junction models • ACM=10-13 uses the Berkeley junction models Level 49 ACM defaults to 0				

A method to specify the diffusion capacitance geometry

3.2 HDIF:

HDIF

heavy diffusion length specified in the model card

length of heavily doped diffusion, used for Area and perimeter calculation in ACM

3.3 XL & XW:

XL	m	0	Length bias accounts for the masking and etching effects
XW	m	0	Width bias accounts for the masking and etching effects

Calculating Effective Length and Width for AC Gate Capacitance

For some MOS processes and parameter extraction method. AC analysis might need different L_{eff} and W_{eff} values than for DC analysis. For AC gate capacitance calculations, substitute the LDAC and WDAC model parameters for LD and WD in the L_{eff} and W_{eff} calculations. You can use LD and WD in L_{eff} and W_{eff} calculations for DC current.

To use LDAC and WDAC, enter XL, LD, LDAC, XW, WD, and WDAC in the **.MODEL** statement. The model uses the following equations for DC current calculations.

$$L_{eff} = L + XL - 2 \cdot LD$$

$$W_{eff} = W + XW - 2 \cdot WD$$

3.4 CJGATE:

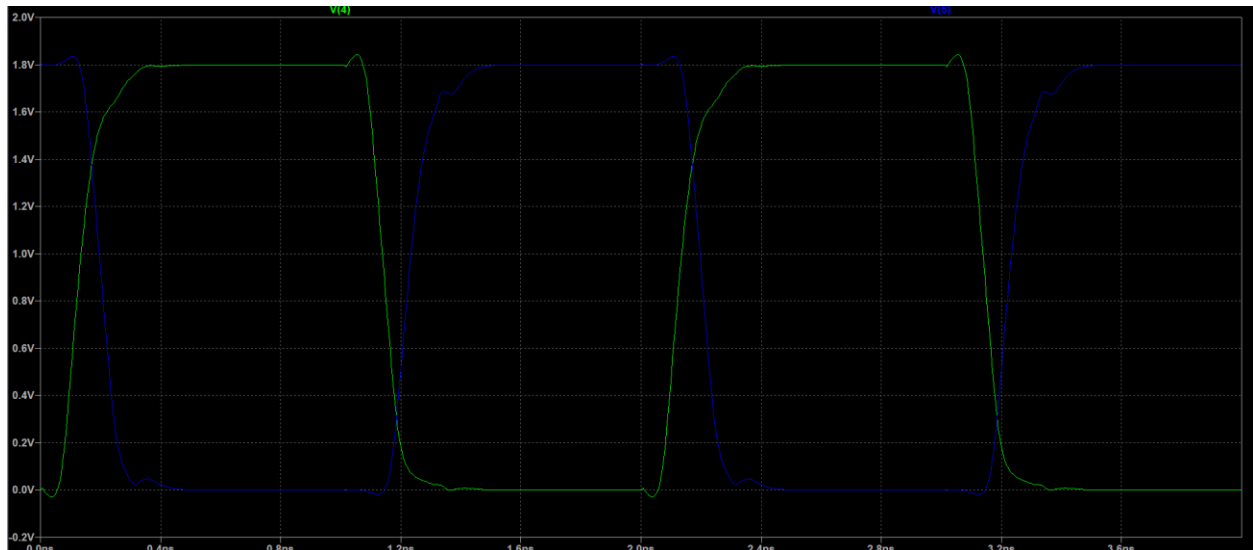
CJGATE	F/m	CJSW	No	Zero-bias gate-edge sidewall bulk junction capacitance (HSPICE-specific) (use only if ACM=3)
--------	-----	------	----	--

Using an ACM=3 MOS Diode

Use ACM=3 to properly model MOS diodes of stacked devices. You can also use the CJGATE parameter to model the drain and source periphery capacitances separately, along the gate edge. Therefore, the PD and PS calculations do not include the gate periphery length. CJGATE defaults to the CJSW value, which in turn defaults to 0.

4 CREATE “CMOS_INV_TRAN_FO4.CIR”. THIS SHOULD SIMULATE A FANOUT-OF-4 (FO4) CHAIN OF FIVE CMOS INVERTERS USING THE “EE214B_HSPICE.SP” MODEL

```
** Circuit Description **
.param mult =1
.subckt inv 1 2 3
M1 3 1 2 2 pch W=20u L=180n m={mult}
M2 3 1 0 0 nch W=10u L=180n m={mult}
.ends inv
VDD 2 0 DC 1.8V
Vin 1 0 PULSE(0 1.8 0 10p 10p 1n 2n)
x1 1 2 3 inv params: mult=1
x2 3 2 4 inv params: mult={4**1}
x3 4 2 5 inv params: mult={4**2}
x4 5 2 6 inv params: mult={4**3}
x5 6 2 7 inv params: mult={4**4}
** Analysis Requests **
.TRAN 5p 4n
*delay defined from 50% of input to 50% of output
.MEAS TRAN Tphl TRIG when v(4) =0.9 cross=1 TARG when v(5)=0.9 cross=1
.MEAS TRAN Tplh TRIG when v(4) =0.9 cross=2 TARG when v(5)=0.9 cross=2
.END
```



5 MEASURE THE FO4 DELAY FOR THE MIDDLE INVERTER IN THE PREVIOUS TESTBENCH. CALCULATE FO4 DELAY (IN PS) DIVIDED BY LAMBDA (IN NM), WHERE LAMBDA = LMIN / 2. THIS RATIO IS USEFUL TO GET A QUICK ESTIMATE OF THE DELAY IN ANY TECHNOLOGY. HOW MUCH FO4 DELAY DO YOU EXPECT FOR A 65NM TECHNOLOGY?

```

SPICE Output Log: D:\spice\LAB3\cmos_inv_tran_fo4.cir.log
LTspice 24.0.12 for Windows
Circuit: CMOS inverter tran analysis
Start Time: Wed Aug 7 01:56:26 2024
Ignoring BSIM parameter ACM
Ignoring BSIM parameter HDIF
Ignoring BSIM parameter XL
Ignoring BSIM parameter XW
Ignoring BSIM parameter ACM
Ignoring BSIM parameter HDIF
Ignoring BSIM parameter XL
Ignoring BSIM parameter XW
Error on line 8 : .model pch pmos ( acm = 3 hdif = 0.32e-6 level = 8 version = 3.3 tnom = 27 tox = 4.1e-9 xj = 1e-7 nch = 4.1589e17 vth0 = -0.3
* Unrecognized parameter "cjgate" -- ignored
Error on line 7 : .model nch nmos ( acm = 3 hdif = 0.32e-6 level = 8 version = 3.3 tnom = 27 tox = 4.1e-9 xj = 1e-7 nch = 2.3549e17 vth0 = 0.36
* Unrecognized parameter "cjgate" -- ignored
solver = Normal
Maximum thread count: 12
tnom = 27
temp = 27
method = modified trap
Warning: Pd = 0 is less than W.
Warning: Ps = 0 is less than W.
Warning: Pd = 0 is less than W.
Warning: Ps = 0 is less than W.
Direct Newton iteration for .op point succeeded.

tphl=7.43535e-11 FROM 1.28055e-10 TO 2.02408e-10
tplh=8.17494e-11 FROM 1.14495e-09 TO 1.2267e-09

Total elapsed time: 0.198 seconds.

```

```

tphl=7.43535e-11 FROM 1.28055e-10 TO 2.02408e-10
tplh=8.17494e-11 FROM 1.14495e-09 TO 1.2267e-09

```

$$T_{delay} = \frac{t_{phl} + t_{plh}}{2} = 78.05ps$$

$$\frac{T_{delay}}{\text{lambda}} = \frac{78.05}{90} = 0.867 \text{ ps/nm}$$

For 65nm:

$$T_{delay_{65nm}} = 0.867 * 32.5 = 28.18 \text{ ps}$$